

Week 4- What more to check in EDA (Pitfalls)?

Dr. Smitha K G

Senior Lecturer,

College of Computing and Data Science,
Nanyang technological University

Target Leakage & Target Validity

Need to check

- Any feature derived *directly or indirectly* from the target
- Features using future information (post outcome feature)

Why critical?

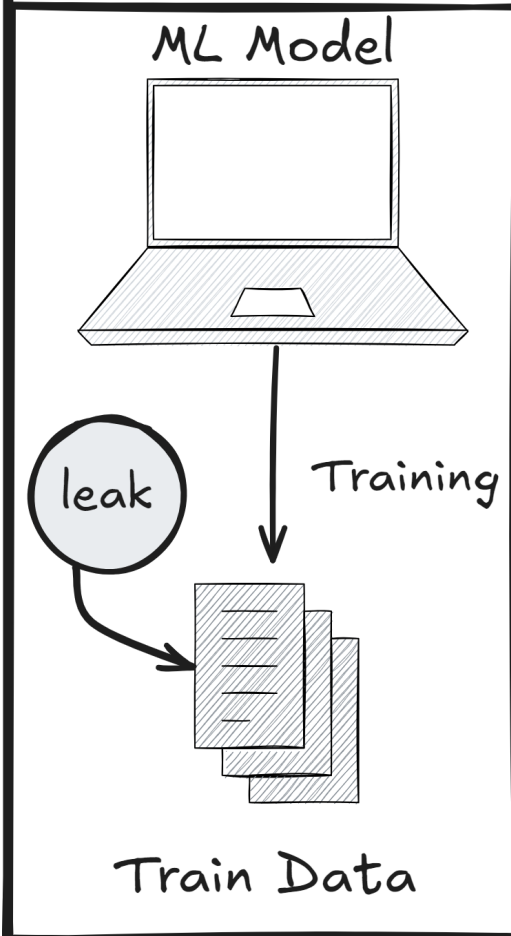
- Produces unrealistically high accuracy (overfitting in training data)
- Fails catastrophically in deployment

Classic examples

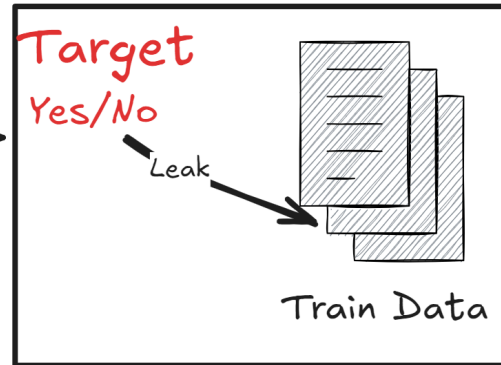
- FinalGrade used to predict Pass/Fail

Data leakage happens when information from outside the training dataset is used to train the model, giving it access to information it shouldn't have during the training phase.

Data Leakage Event



Target Leakage

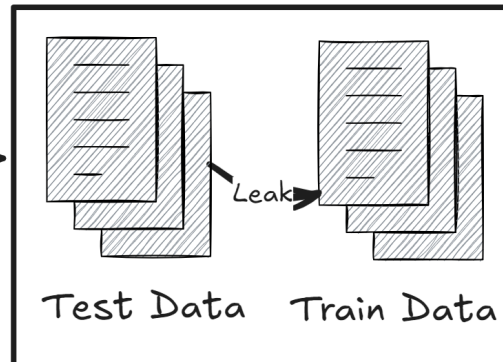


Occurs when the model has access to the target variable or information that would not be available at prediction time

Effect of Data Leakage

1. Overfitting to Training Data
2. Inflated Performance Metrics
3. Misleading Insights
4. Wasted Resources

Train-Test Contamination



Happens when information from the test set gets included in the training process, where the same sample appears in both training and test sets

Common Data Leakage Case

1. Including Future Data in Time-Series Predictions
2. Feature Scaling and Preprocessing Before Splitting
3. Target Leakage Through Derived Features
4. Using the Test Set to Tune the Model

Correlated Feature

Exists before decision

Predictive signal

Legitimate

Post-Outcome Feature

Exists after outcome

Outcome proxy

Illegitimate

Amount recovered is a post-outcome feature because its value is determined only after the borrower has either defaulted or repaid, making it causally dependent on the target variable and unavailable at prediction time

Data Representativeness & Sampling Bias

Need to check

- Who/what is missing from the dataset?
- How was the data collected and does the dataset represent the real population?

Why critical?

- Models generalize only to the population they see
- Leads to systemic bias and unfair predictions and this Invalidates evaluation metrics

Examples

- Loan datasets with only approved applicants
- Medical datasets from a single hospital
- Student datasets from only high-performing institutions

**Class balance $\neq$
population balance.**

Temporal Leakage / Ignoring Time- introduces concept drift

Need to check

- Time dependency in features or labels
- Whether feature–target relationships change over time
- Whether train/test splits respect time

A model that ignores time assumes the world never changes.

Why critical?

- Random splits give false confidence
- Models degrade rapidly after deployment

Examples

- Student performance across academic years
- Customer behavior pre- and post-policy change

TIER 1 — Model-Invalidating Risks

(If ignored → results are fundamentally wrong)

Failure	Looks Good Because	Fails Because
Target leakage	Future info inflates accuracy	Impossible at inference
Sampling bias	Clean metrics on biased data	Wrong population
Temporal leakage	Random split hides drift	World changes

Tier-1 failures cannot be fixed by better models, better tuning, or more data. They require redesigning the dataset itself.

1. The target is leak-free and meaningful
2. The data represents the real population
3. The model will still work when time moves forward

Multicollinearity

What to check

- Strong correlations among predictors
- Variance Inflation Factor ($VIF > 5-10$)
- Redundant engineered features
- Semantically overlapping variables

Why it is critical

- Feature importance becomes unreliable
- Causal and policy interpretations become invalid

Examples

- Predict house price using: 'HouseAge' 'YearsSinceRemodel' as these features are strongly correlated.

Multicollinearity does not kill accuracy — it kills explanations

Effects of Multicollinearity

When two predictors are **highly correlated**, the regression model:

- Cannot uniquely decide **which feature deserves credit**
- Small noise → large coefficient changes
- Coefficients may flip sign even if the true relationship is known
- Interpretation contradicts domain logic
- The model is still minimizing error correctly — but explanations become unstable.
- It is good to : Drop redundant features and combine semantically similar features

Label Noise/Quality

What to check

- Incorrect, inconsistent (same input gets different labels depending on annotator, time and context) or proxy labels
- Eg: Self reported survey labels- On a scale of 1–10, how stressed are you?

Why it is critical

- Makes decision boundary fuzzy
- Regularization cannot fix wrong labels
- Model confidence is reduced as it fails real learning objectives

Examples

- Human-annotated sentiment analysis, Eg:-Classifying reviews as Positive / Neutral / Negative (“The phone is amazing, but the battery dies in 3 hours.”)
- One can say it to be Positive (phone amazing), another neutral (one positive one negative), another negative(as it dies in 3 hours) – Subjective interpretation and even fatigue

Outlier semantics

What is it

- Outlier semantics means understanding what an outlier represents in the real world, not just whether it is statistically extreme.
- An outlier is not automatically a noise or error— it may be a valid, rare, or important population or a distinct subgroup.

Why it is critical

- Model learns only “average” or “ dominant” behavior once outlier is removed.
- Sometimes outliers are our major signals to learn (understanding a rare medical condition)

Examples

- Rare medical condition- Genetic disorder for 1% of population
- As they are statistically and outlier, the model may remove that and never learns the disease patterns.

TIER 2 — Interpretation and Stability Risks

(Model may appears to work, but its explanations, stability, or fairness are unreliable)

Failure	Looks Good Because	Fails Because
Multicollinearity	- High accuracy - Training & validation scores look stable	- Individual feature effects become unreliable - Coefficients may flip sign or magnitude with small data changes
Label Noise / Poor Label Quality	Model trains successfully without errors	- Model learns noise or annotator bias instead of signal - Errors persist even with more data or tuning
Outlier Semantics Ignored	- Metrics improve after outlier removal - Distributions look cleaner and more “normal”	- Rare but important populations are erased - Decision boundaries shift toward majority behavior

Tier-2 risks do not reduce apparent performance immediately; they undermine trust, interpretability, and robustness over time.

TIER 3 — Model compatibility and Optimization

(Model works but not optimally- performance degraded)

Failure	Looks Good Because	Fails Because
Feature Scaling and Distribution shape (skew)	1. Model trains without errors 2. Metrics may look acceptable on small datasets	1. Distance-based models distort similarity due to scale imbalance 2. Heavy-tailed/skewed features dominate gradients
Dimensionality and feature redundancy	Adding features initially improves training performance	1. Training time, memory usage, and inference cost increase unnecessarily 2. Redundant features inflate variance and overfitting risk

Tier-3 risks are often mistaken for “model weakness” when the root cause lies in preprocessing and representation.